



# BRDRUM

Rhythm Composer - 001

## BRDRUM Manual

From your first pattern to the exported stem. Every screen captured from the device, with the path marked above it.



### WHAT IS IN HERE

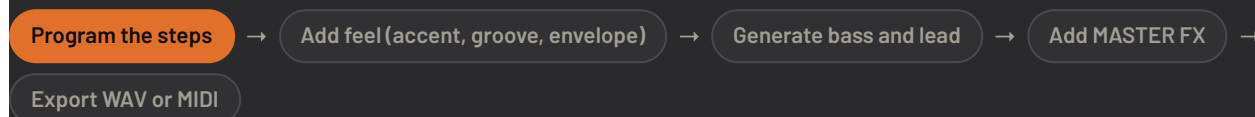
1. **How BRDRUM works** — what it is, and the map of the panel
2. **Getting ready** — installing, activating your license, choosing your sounds
3. **Day to day** — programming, adding feel, FILL, MIDI GEN, MASTER FX and EXPORT
4. **Every control, what it does** — quick reference
5. **When something goes wrong** — what to do

# How BRDRUM works

A 6 instrument drum machine, with a 16 or 32 step sequencer, a bass and lead MIDI generator, master FX and stem export.

You program the steps for each instrument, bring the pattern to life with accent, groove and each row's envelopes, add the master FX, and export stems or MIDI ready for your DAW. MIDI GEN writes bass and lead lines in the scale you pick, ready to drag straight into your session.

## From a pattern to a stem



## The map of the panel



- 1 **Header.** SAMPLER, the LICENSE indicator and SIGN OUT, the device name, TONE X, RAND and EXPORT.
- 2 **Rows.** The 6 instruments, KICK, SNARE, CLAP, CHAT, OHAT and TOM. Each with MUTE, SOLO, CLEAR, a per row RAND and the DECAY, RELEASE, TUNE, VOL, PAN and NOISE knobs. KICK also has KEY and PUNCH.
- 3 **Grid.** The 16 or 32 steps for each row, in 4 color groups, plus REVERB, DELAY and the ECHO key for each instrument. The strip the header's RAND controls (the grid header down to GROOVE and ACCENT) sits on a subtle raised step, to mark how far it reaches.
- 4 **ACCENT and GROOVE.** The per step ACCENT slider, and GROOVE, now in 9 families, with the AMOUNT knob.
- 5 **MIDI GEN.** KEY, SCALE, BARS, and the BASS and LEAD buttons.
- 6 **FILL, Pal and MASTER.** The Pal character, the MASTER knob, and the FILL panel, with its target, RATE, OFFBEAT and VOLUME.
- 7 **MASTER FX.** SATURATION, MULTIBAND and ECHO on the master bus.
- 8 **Footer.** STOPPED or PLAYING, and the last thing the device did.

# Getting ready

Three things before your first pattern. You do these once, and only touch them again if you switch machines or sounds.

## 1 Install

Installer → VST3 and AU

The installer places BRDRUM in your system's standard plugin folders, as VST3 and, on Mac, also as AU, and writes the factory samples to the system's factory folder.

| SYSTEM  | WHERE IT LIVES                                                                                                                                                        |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| macOS   | VST3 in <code>/Library/Audio/Plug-Ins/VST3</code> , AU in <code>/Library/Audio/Plug-Ins/Components</code> .<br>Factory samples in <code>/Users/Shared/BRDRUM</code> . |
| Windows | VST3 in <code>Common Files\VST3\BRDRUM.vst3</code> . Factory samples in <code>C:\ProgramData\BRDRUM</code> .                                                          |

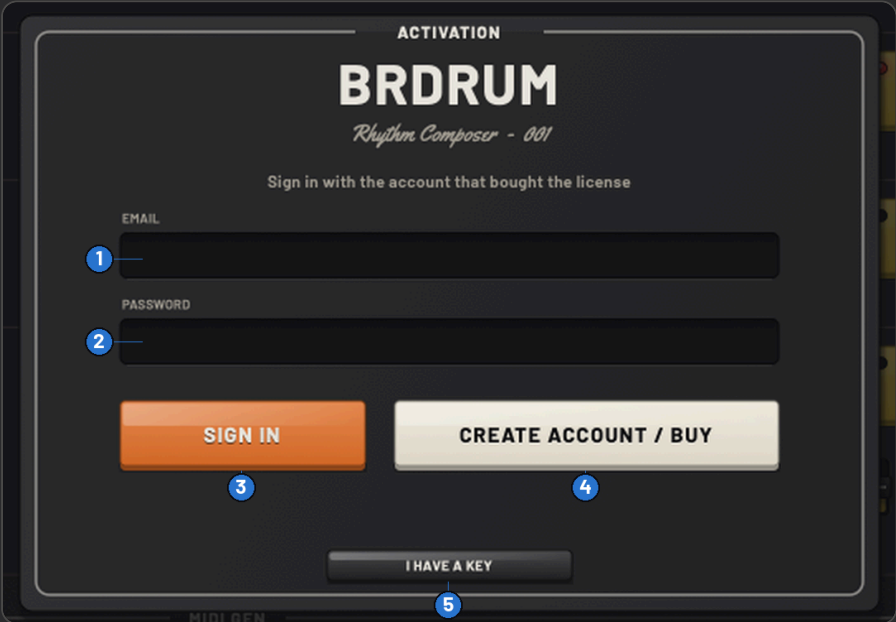
### Rescan after installing

Once the plugin is installed, rescan plugins in your DAW (or open your DAW for the first time after installing) for BRDRUM to show up in the list.

## 2 Activate your license

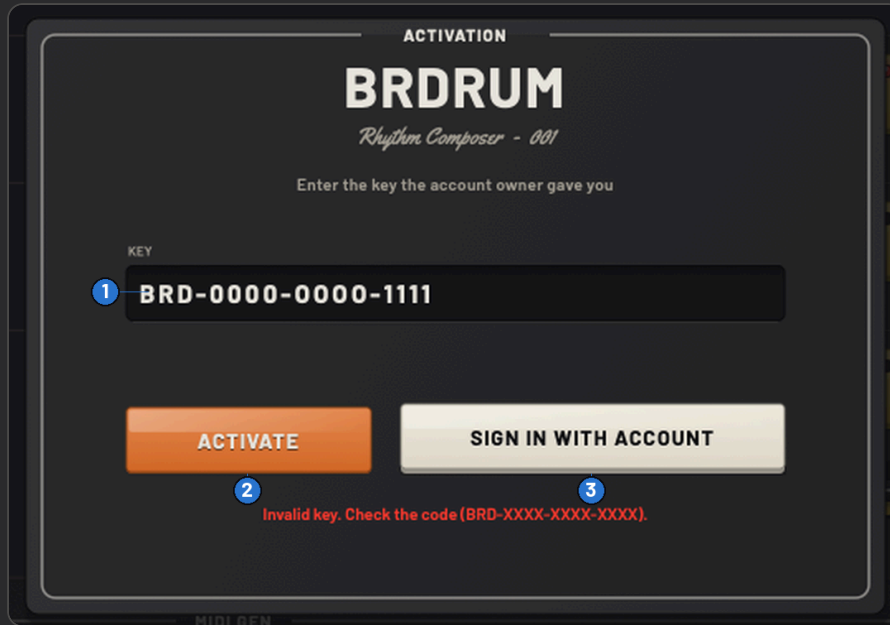
ACTIVATION screen, when opened without a license

Without a license, the header shows **NO LICENSE** and BRDRUM opens with the ACTIVATION screen over the panel. You sign in with the account that bought the license, or with a key the account owner gave you.

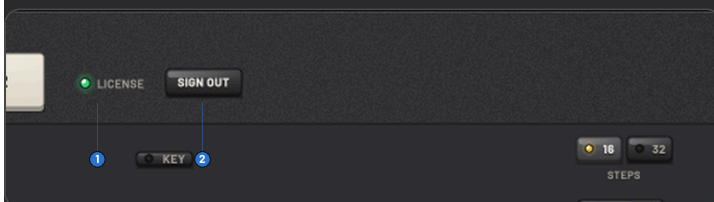


- 1 **EMAIL.** The email of the account that bought the license.
- 2 **PASSWORD.** That account's password.
- 3 **SIGN IN.** Activates this device on that account.
- 4 **CREATE ACCOUNT / BUY.** Takes you to the site, for anyone who does not have a license yet.

- 5 **I HAVE A KEY.** Opens the key screen, for anyone who got a key from the account owner instead of an email and password.



- 1 **KEY.** The code, in the format `BRD-0000-0000-0000`, that the account owner gave you.
- 2 **ACTIVATE.** Activates this device with that key. A wrong key shows the warning in red: `Invalid key. Check the code (BRD-XXXX-XXXX-XXXX).`
- 3 **SIGN IN WITH ACCOUNT.** Goes back to the email and password screen.



- 1 **LICENSE.** With the license active, the LED turns green and the text shows the account (email) or the key in use, with the SEATS in use.
- 2 **SIGN OUT.** Releases this device's access. Activated by account, it also frees the seat on the server for another device. Activated by key, it only signs out this device, the seat stays as it was.

### Price and purchase

Price and payment live on the site. This manual does not repeat that number here, because it changes.

## 3 Choose your sounds

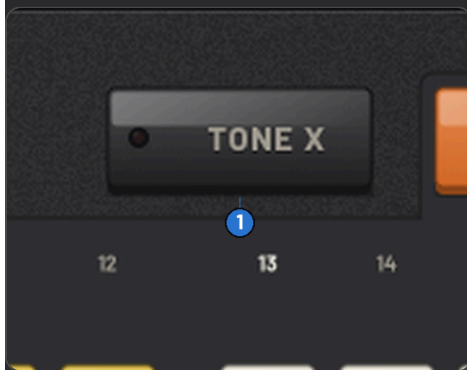
SAMPLER, and the TONE X key

SAMPLER opens BRDRUM's sound folder on your system. It adds together two groups of samples: the factory ones that came with the installer, and your own, which you drag into the right instrument's folder. When it plays, BRDRUM draws from both.



- 1 **SAMPLER.** Opens the user sound folder on your system. Drop a `.wav` into the right instrument's folder to add it to the draw.

**TONE X** is a header key that extends each instrument's tuning range. For it to apply to a row, two keys have to be on at the same time: the header one, and that row's LED, right next to TUNE. With the header key off, the row LEDs make no difference to the sound at all.



- 1 **TONE X.** Turns on the extended tuning range for the whole panel.



- 1 **Row LED.** Each row decides whether to use the extended range, with this LED next to TUNE.

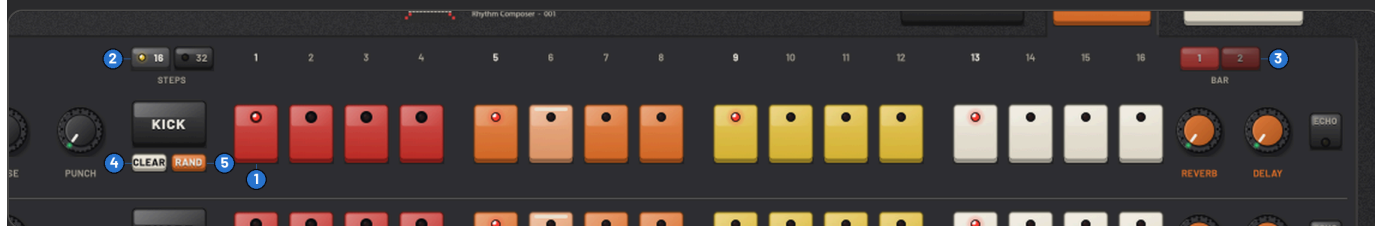
## Day to day

With setup done, it is time to program a pattern and take it to a stem or MIDI file. That is six steps on the panel.

### 4 Program a pattern

The instrument rows, and the step grid

Each row is an instrument. Click the grid squares to turn a step for that row on or off. The steps show up in 4 color groups, four at a time, so you can count the bar at a glance.



- 1 **Step keys.** One click turns that row's step on, another click turns it off. The 4 color groups mark the bar's 4 beats.
- 2 **STEPS.** Switches between 16 and 32 steps. With 32, the pattern becomes 2 bars.
- 3 **BAR.** Only shows up with 32 steps on. Picks which of the 2 bars you are editing, and shows the right numbers above the grid, 1 to 16 or 17 to 32. Switching back to 16 does not erase bar 2, it just hides it.
- 4 **CLEAR.** Twice. The first click asks for confirmation in the status bar for 3 seconds, the second clears that row's steps, accents and knobs. The loaded sample, MUTE, SOLO and TONE X stay as they were.
- 5 **Per row RAND.** Next to CLEAR. Rolls only that row's grid, over the length you are playing, with its own recipe per instrument (KICK rolls offbeats, SNARE and CLAP roll fills, OHAT varies). Does not change the sample or the knobs.

In the column to the left of each row (outside this crop, see the panel map in Part 1), **MUTE** silences only that row and **SOLO** silences the others that are not soloed. You can put more than one instrument in solo at the same time. With Cmd (Mac) or Ctrl (Windows) held down, clicking SOLO isolates only that row and clears solo on the rest.

The header's **RAND** key is different from each row's RAND: it rolls the whole grid, the samples, every channel's knobs and the ACCENT, all at once. That is why the raised strip on the panel map (Part 1) marks how far it reaches, from the grid header down to GROOVE and ACCENT.

### 5 Bring the pattern to life

ACCENT, GROOVE, each row's knobs, and ECHO

With the pattern programmed, each row's controls and the groove take it out of the grid. On KICK, KEY and PUNCH deserve special attention.



- 1 **KEY and the note.** KICK only. Off, TUNE works normally and shows the note only while you turn the knob. On, the kick's tuning follows the root note picked in MIDI GEN's KEY, TUNE starts moving in semitones, and the note stays written the whole time.
- 2 **DECAY, RELEASE, TUNE, VOL, PAN and NOISE.** That instrument's envelope, tuning, volume, pan and amount of noise.
- 3 **PUNCH.** KICK only. From 0 to 100%, adds a faster attack and a boost at the kick's fundamental frequency. At 0% nothing about the sound changes. The boost follows whatever pitch is currently playing, so it keeps working even with KEY on.



- 1 **REVERB and DELAY.** That row's send to the master reverb and delay.
- 2 **ECHO.** A key on each row. Off on every row, the MASTER FX ECHO picks up every instrument's sound. On for one or more, only those feed the echo, the others stay dry in it, but still pass through that row's normal DELAY.



- 1 **ACCENT.** One slider per step. Marks which steps hit harder.
- 2 **GROOVE.** 9 families of rhythmic swing: Straight, Random, Ableton Swing 8, Ableton Swing 16, MPC 16 Swing, SP-1200 Swing, TR-909 Shuffle, FL Swing and HARD Swing.
- 3 **AMOUNT.** The intensity within the chosen family. The legend shows the real value, for example MPC 62%, SHUFFLE 4 or FL 75%. An old project opens already on the family that matches the groove it used.

## 6 FILL

Right column, below Pal

FILL plays an automatic variation on a chosen instrument, at the interval you set. It is the fill that plays itself, without you programming step by step.

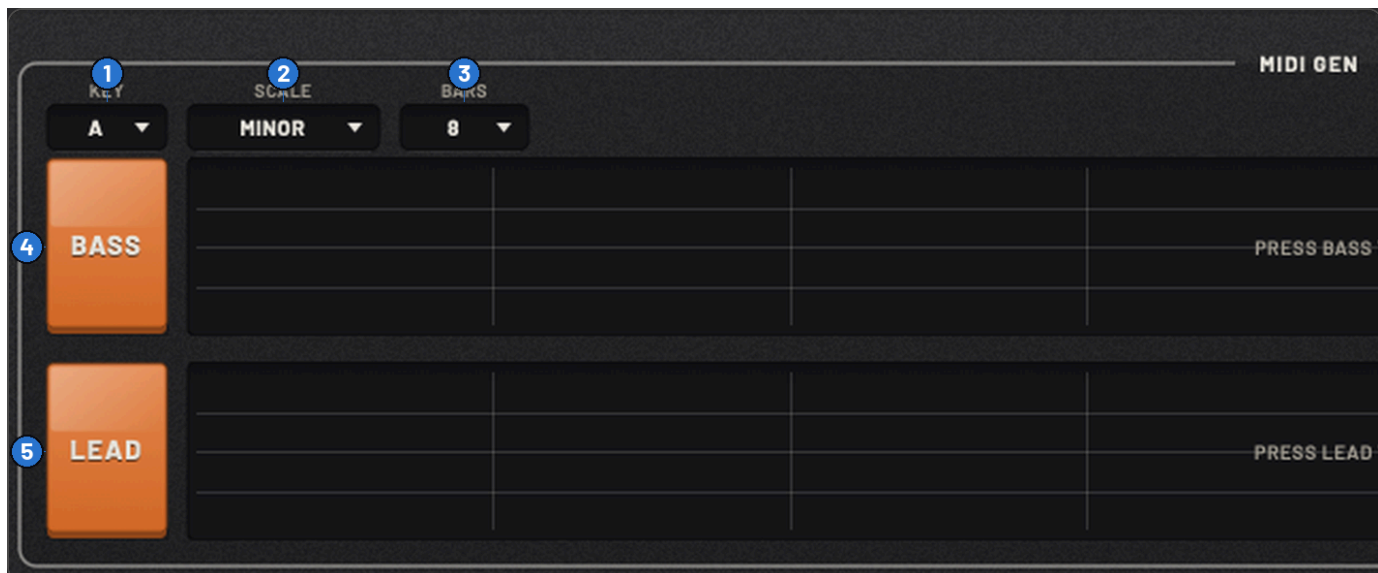


- 1 **Target.** The instrument that gets the fill. SNARE by default.
- 2 **RATE.** A bipolar knob, with 0 at the center. Left to right: 2 · 1 · 3/4 · 0 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32. At 0, the fill is off.
- 3 **OFFBEAT.** A key separate from RATE. Shifts the fill half a step, onto the offbeat.
- 4 **VOLUME.** The fill's volume.

## 7 MIDI GEN

Center of the panel, below the grid

MIDI GEN writes a bass line and a lead line in the scale you picked. You do not have to program note by note: generate, listen, and drag it into your DAW if you like it. Empty, each lane shows **PRESS BASS TO GENERATE** or **PRESS LEAD TO GENERATE**.



- 1 **KEY.** The scale's root note. It is the same note KICK's KEY follows.
- 2 **SCALE.** The type of scale.
- 3 **BARS.** 8 or 16 bar phrase.
- 4 **BASS.** Generates a new bass line in that root note and scale.
- 5 **LEAD.** Generates a new lead line, matched to the bass.

With notes on the lane, click it and drag it onto a track in your DAW: the label `DRAG TO YOUR DAW` shows up in the corner once there is something to drag. Out comes a MIDI file with that whole phrase.

#### Where the MIDI files go

BRDRUM keeps a copy of what you drag out in `BRDRUM/MIDI/Bass` and `BRDRUM/MIDI/Lead`, inside your system's sound folder.

## 8 MASTER FX

Bottom strip of the panel, below MIDI GEN

Three effects in series on the master bus: SATURATION, MULTIBAND and ECHO. Each one turns on with its ON key, and opens a larger panel with EDIT.





- 1 **SATURATION ON.** Turns on the master saturation.
- 2 **Type.** The saturation curve shape, Analog Clip by default.
- 3 **DRIVE, OUTPUT, DRY/WET.** How much it saturates, the output level, and the blend with the clean signal.
- 4 **EDIT.** Opens the full saturation panel.
- 5 **MULTIBAND ON.** Turns on the 3 band compressor.
- 6 **AMOUNT, TIME, OUTPUT, LOW, MID, HIGH.** Overall intensity, response time, output, and each band's gain.
- 7 **EDIT.** Opens the full multiband panel.
- 8 **ECHO SYNC.** Turns on echo synced to the host tempo. Without SYNC, TIME is free, in milliseconds. With SYNC, TIME becomes a time division that grows clockwise around the knob: `0 · 1/32 · 1/16 · 1/8 · 1/4 · 1/2 · 3/4 · 1 · 2`, with 0 (off) at the left end. Switching divisions does not click.
- 9 **TIME, FEEDBACK, INPUT, FILTER, MOD%, DRY/WET.** Echo time, how much it repeats, how much signal goes in, the filter, modulation and blend.

#### SATURATION's EDIT panel



- 1 **Curve.** Shows the transfer curve for the chosen type, with the current drive.
- 2 **COLOR.** Pre and post emphasis for lows and highs, with AMT LO, AMT HI, FREQ and WIDTH.
- 3 **CLIP.** An extra limiter after the curve, with No Clip, Soft and Hard options.
- 4 **WAVESHAPER.** Drive, Linear, Curve, Damp, Depth and Period. Only apply when the chosen type is Waveshaper.

5 **FECHAR** ("close"). Goes back to the MASTER FX strip.

## MULTIBAND's EDIT panel



1 **SOFT KNEE and RMS**. Global keys, for a smoother curve and average based detection instead of peak.

2 **PRESET**. Padrão (default), or OTT, for a more aggressive multiband.

3 **HIGH, MID, LOW**. Turns each band on with ON, isolates it with the S solo key, and sets where one band ends and the next begins with the HI/MID and MID/LO knobs.

4 **INPUT through OUTPUT**. Each band works as its own compressor, with input, two ratio and threshold stages, attack, release and output.

5 **GR**. Each band's gain reduction meter, in real time.

6 **FECHAR**. Goes back to the MASTER FX strip.

## 9 **EXPORT: stems or MIDI**

EXPORT key, in the header

The EXPORT key asks first what to export. With 32 steps on, you get 2 bars.

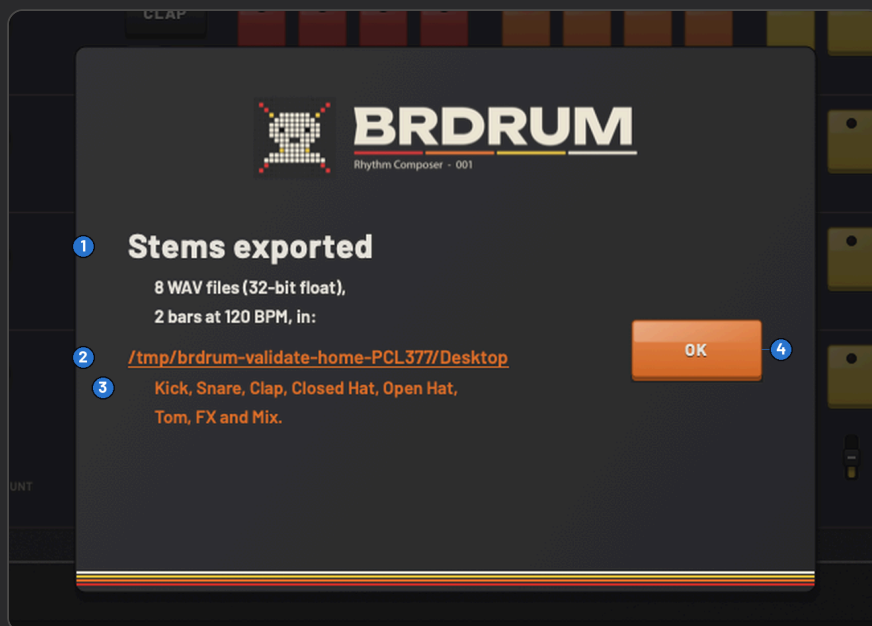


1 **Description**. WAV writes 8 audio stems. MIDI writes the drum pattern, one file per instrument plus one with everything.

2 **WAV**. Pick the folder and write the 8 stems.

3 **MIDI**. Pick the folder and write the 7 .mid files.

4 **Cancel.** Closes without exporting anything.



1 **Confirmation.** How many files, the format, and how many bars, at the host's BPM.

2 **Folder.** The path the files were saved to. Click it to open the folder.

3 **List.** The 8 stems that came out: Kick, Snare, Clap, Closed Hat, Open Hat, Tom, FX and Mix. Each file is named `Instrument BPMBPM.wav`, as 32 bit float WAV.

4 **OK.** Closes the notice.

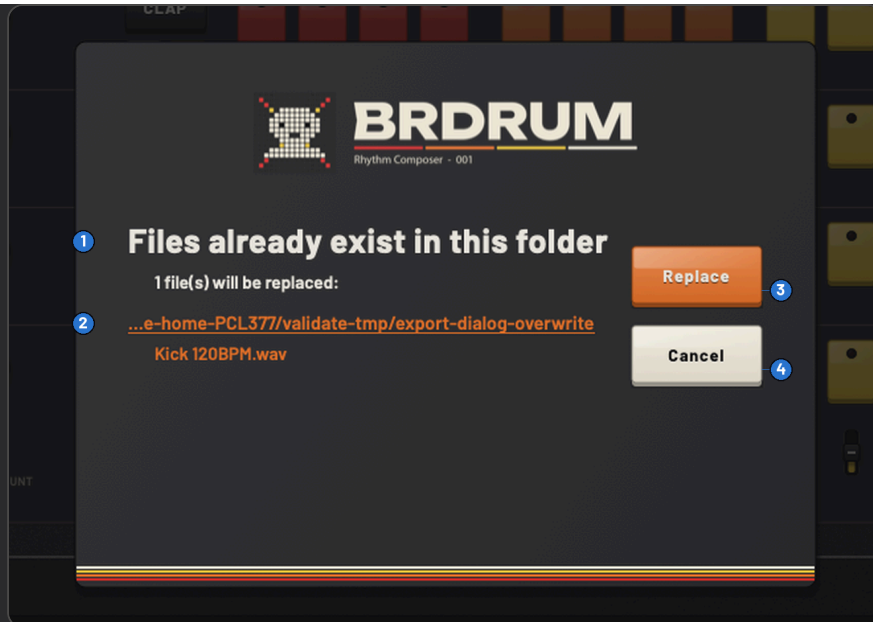


1 **Confirmation.** How many files and how many bars, at the host's BPM.

2 **Folder.** The path the files were saved to.

3 **List.** The 7 .mid files: Kick, Snare, Clap, Closed Hat, Open Hat and Tom, one per instrument, plus Drums with all six rows together. Each MIDI track comes out named after the stem, ACCENT becomes note velocity, and GROOVE's swing goes into the note positions.

4 **OK.** Closes the notice.



- 1 **Warning.** Files with these names already exist in the chosen folder.
- 2 **List.** Which file or files will be overwritten.
- 3 **Replace.** Writes over the old files.
- 4 **Cancel.** Exits without exporting anything.

# Every control, what it does

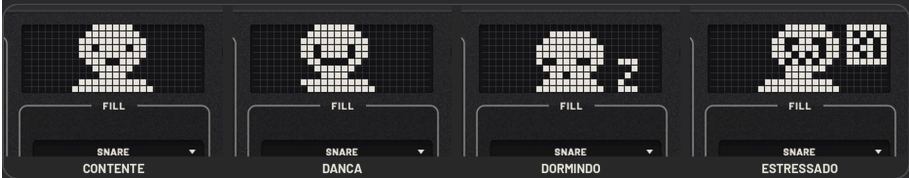
Quick reference. Use it when you know what you want but not where it lives.

| REGION            | WHAT IS THERE                                                                                                     |
|-------------------|-------------------------------------------------------------------------------------------------------------------|
| Header            | SAMPLER, LICENSE and SIGN OUT, the device name, TONE X, RAND and EXPORT.                                          |
| Rows              | The 6 instruments, with MUTE, SOLO, CLEAR, a per row RAND, the envelope and tuning knobs, REVERB, DELAY and ECHO. |
| Grid              | 16 or 32 steps per row, in 4 color groups. The STEPS key (16 32), and BAR (1 2) when in 32.                       |
| ACCENT and GROOVE | One accent slider per step, and GROOVE in 9 families with AMOUNT.                                                 |
| MIDI GEN          | KEY, SCALE, BARS, and the BASS and LEAD generators, to drag into your DAW.                                        |
| Pal               | The pixel screen character. Reacts to what you do and looks straight at you with the mouse over it.               |
| FILL              | Target, bipolar RATE, OFFBEAT and VOLUME, for the automatic fill.                                                 |
| MASTER FX         | SATURATION, MULTIBAND and ECHO on the master, each with an EDIT panel.                                            |
| EXPORT            | WAV (8 stems) or MIDI (7 files), warning before it overwrites anything.                                           |

## KICK, the two exclusive controls

| CONTROL | WHAT IT DOES                                                                                                                                     |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| KEY     | Tunes the kick to MIDI GEN's root note. The note stays written the whole time, and TUNE starts moving in semitones.                              |
| PUNCH   | Faster attack and a boost at the kick's fundamental frequency, following whatever pitch is currently playing (even with KEY on). From 0 to 100%. |

## Pal, the character



Pal reacts to what happens on the panel. Contente (happy) is the normal state, and it follows your mouse, looking straight at you when the mouse is over it. Danca (dancing) when the DAW is playing. Dormindo (sleeping) when nobody has touched anything for a while. Estressado (stressed) when the overall level is very low, on the master or on the channels.

# When something goes wrong

| WHAT YOU SEE                              | WHAT TO DO                                                                                                                                                             |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No sound                                  | Check LICENSE in the header. Then check whether an instrument is on MUTE, or another one on SOLO is muting the rest. Check that row's VOL and the MASTER.              |
| The device opens on the ACTIVATION screen | This device's license is not active. Sign in with your account or with a key, as in Part 2.                                                                            |
| Invalid key, or no SEATS left             | Check the key's format, <code>BRD-0000-0000-0000</code> . If every SEAT on the account is already in use, click SIGN OUT on another device before activating this one. |
| EXPORT came out incomplete                | Check that the chosen folder exists and that you have write permission on it. Export again.                                                                            |
| The KICK note does not show up            | With KEY off, the note only shows while you turn TUNE. Turn KEY on to keep the note written the whole time.                                                            |
| Pal looks stuck                           | It falls asleep if nobody touches anything for a while. Move the mouse or click something to wake it up.                                                               |

## No sample means no sound, even with the step on

If you cleared the user sample folder, BRDRUM still adds in the factory ones. With no sample in either folder for that instrument, the step turns on but does not play.